

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 3.3: Excretion and Homeostasis in Animals — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 3.3: Excretion and Homeostasis in Animals
Driving Question	DRIVING QUESTION: How do different animals maintain a balanced internal environment and eliminate toxic wastes despite drastic changes in their external conditions?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Same Hot Afternoon, Different Bodies: Launching the Excretion and Homeostasis Puzzle	Students compared how a Kenyan dog, a lizard basking on a rock near Lake Victoria, a tilapia from the lake, a goat and a person behave and look on the same hot afternoon (panting, sweating, basking, shade-seeking). They also compared the wastes of these animals: dilute urine in tilapia, concentrated urine in the goat, and white semi-solid uric acid paste in lizards and chickens. Teacher note: Use WAIT TIME of 5 seconds after asking 'Why do you think the lizard's waste looks so different from the tilapia's?' and cold-call 4 students to share predictions before giving any explanation. Record all predictions on the Driving Question Board (DQB) without correcting them.	The type of nitrogenous waste an animal makes is linked to how much water is available to it. Animals with plenty of water (freshwater fish such as tilapia) excrete ammonia, which is very toxic and needs a lot of water to dilute it. Mammals such as goats and humans make urea, which is less toxic and can be stored briefly and excreted in a moderate volume of water. Birds, lizards and insects, which conserve water, make uric acid, which is almost insoluble and non-toxic and is excreted as a paste with very little water loss. Excretion is the removal of toxic metabolic wastes, and homeostasis is the maintenance of a stable internal environment. Teacher note: Accept the idea that different habitats lead to different strategies, and keep unresolved questions (e.g. 'How does the dog cool down?') on the DQB for later lessons.	Model answer: Animals in different environments face different challenges, so they have different ways of removing nitrogenous wastes. A tilapia is surrounded by water so it can flush out ammonia easily, whereas a lizard in the dry Rift Valley must save water and so excretes uric acid. This begins to answer the driving question because the internal environment is kept safe by matching the waste type to water availability. Pedagogical note: Common misconception to watch for is that 'excretion' means only 'urination' or the same as egestion (removal of undigested food). Clarify that excretion removes wastes made by cells. Students should draw an initial model with the animals, their wastes and the environments, to be revised in later lessons.
Lesson 2 Inside the Goat Kidney: Cortex, Medulla, Pelvis and Nephrons	Students dissected or examined a fresh goat kidney (sourced from a local butcher in the market) or a labelled diagram or photographs. They cut it longitudinally and observed the dark outer cortex, the paler	The mammalian kidney is organised into regions with different roles. The cortex contains many nephrons where blood is filtered (the glomerulus and Bowman's capsule, with the convoluted tubules), the	Model answer: The kidney is built so that blood is filtered in the cortex, and the water-saving work is done in the medulla. Millions of nephrons give a large surface area for filtering the blood. Linking to the

	inner medulla with its pyramids, the funnel-shaped pelvis, and the ureter leaving the kidney. They then examined slides, diagrams or a model of a nephron. Teacher note: Safety first. Students must wear gloves, wash hands with soap after, and handle the sharp tools only under supervision. Ask 'Where do you think blood enters and where does urine leave?' and give 5 seconds of WAIT TIME before cold-calling 3 students.	medulla contains parts of the nephron such as the loops of Henle and collecting ducts that concentrate urine, and the pelvis collects urine and passes it to the ureter. The renal artery brings blood with urea to the kidney, and the renal vein takes away blood with less urea. The nephron is the functional unit of the kidney. Teacher note: Check that students can label a nephron: Bowman's capsule, glomerulus, proximal convoluted tubule, loop of Henle, distal convoluted tubule and collecting duct.	phenomenon, a goat or a human on a hot afternoon can make concentrated urine because the medulla and its loops of Henle can reabsorb water. Pedagogical note: Students often think the kidney stores urine (it is the bladder) or that urine is made only in the medulla. Have them trace the path of a drop of blood and a drop of urine using arrows on a diagram, and add the structures to the class model and DQB.
Lesson 3 Following the Fluid: How the Nephron Filters Blood and Makes Urine	Students followed the path of fluid through a nephron using a data table of concentrations (glucose, urea, proteins, water, salts) in blood plasma, glomerular filtrate and urine, and a modelling activity using a sieve or cloth filter with a mixture of water, sand and salt or maize flour. They saw that large molecules were held back and small molecules passed through. Teacher note: Ask 'Why is there no glucose in healthy urine?' and use 5 seconds of WAIT TIME. Cold-call 3 students to describe where glucose is lost from the data and where it was reabsorbed.	Urine formation involves ultrafiltration in the glomerulus, where small molecules (water, glucose, amino acids, urea, salts) are forced into Bowman's capsule under high pressure while blood cells and large proteins stay in the blood. This is followed by selective reabsorption, mostly in the proximal convoluted tubule, where all glucose and amino acids and much water and salts are returned to the blood. Water is further reabsorbed in the loop of Henle and collecting duct, and salts are adjusted in the distal tubule. Urea and excess salts and water remain and leave as urine through the ureter, bladder and urethra. Teacher note: High pressure is caused by the afferent vessel being wider than the efferent vessel.	Model answer: Blood is first filtered non-selectively by size and then the body takes back what it needs, leaving the wastes to be excreted. This is how a person who has been sweating in Nairobi traffic can have small volumes of dark, concentrated urine, because more water is reabsorbed. Pedagogical note: Common misconception is that the kidney filters out only wastes, whereas it filters out many useful substances and then reabsorbs them. If glucose or protein appears in urine, this can be a sign of problems such as diabetes or kidney damage. Students update the model with the steps of filtration and reabsorption and add questions to the DQB.
Lesson 4 Liver and Skin: Hidden Waste Processors Behind the Hot Afternoon	Students studied diagrams, a short reading, and data on the liver and skin. They observed a labelled diagram of a section of skin with sweat glands, and compared the composition of sweat and urine. They considered what happens to proteins in a meal of beans or nyama choma, and why sweat is salty. Teacher note: Ask 'Where does the urea come from before it reaches the kidney?' and allow 5 seconds of WAIT TIME. Cold-call 3 students and use their ideas to link the liver, blood and kidney.	The liver deaminates excess amino acids to make urea (the nitrogen part is removed and converted, and the rest is used for energy), breaks down haemoglobin from old red blood cells to make bile pigments such as bilirubin that leave in bile, and processes toxic substances such as alcohol and drugs (detoxification). The skin excretes water, salts and a little urea in sweat, and sweating also helps cool the body. The lungs excrete carbon dioxide and water vapour. Teacher note: Stress that the liver makes urea but the kidney removes it from the blood.	Model answer: Several organs work together in excretion. The liver converts toxic ammonia from amino acid breakdown into urea, which is carried by blood to the kidneys, while the skin loses water and salts as sweat. On a hot afternoon, a footballer or farmer in Kisumu loses a lot of water and salts in sweat, so less urine is produced. Pedagogical note: Students may think sweating exists only for excretion, so link it to temperature regulation, which is developed in Lesson 6. Add the liver and skin to the class model of how wastes are made and removed, and post new questions on the DQB.

Lesson 5 Feedback Loops: How the Body Keeps Water and Blood Glucose Steady	Students analysed graphs and data on blood glucose after a meal of ugali and sukuma wiki, and urine volume after drinking water compared with a hot day with little water. They then built a flow diagram or acted out a role-play with cards for stimulus, receptor, control centre and effector. Teacher note: Ask 'What would happen if blood glucose kept rising and nothing responded?' with 5 seconds of WAIT TIME and cold-call 4 students. Use a thermostat or a gas stove analogy to introduce the loop.	Negative feedback detects a change from a set point and triggers a response that reverses it. ADH (antidiuretic hormone), released by the pituitary gland when the blood is too concentrated, makes the kidney collecting ducts more permeable so more water is reabsorbed and less, more concentrated, urine is produced; when blood is too dilute, less ADH is released and more dilute urine is produced. Blood glucose is regulated by insulin (lowers glucose by promoting uptake and conversion to glycogen in the liver) and glucagon (raises glucose by breaking down glycogen), both from the pancreas. Diabetes mellitus results when this control fails. Teacher note: Insist on the use of the terms stimulus, receptor, coordination centre and effector.	Model answer: Homeostasis works through negative feedback, so the internal environment returns to its normal level after any change. After a person sweats, the blood becomes concentrated, ADH increases and water is conserved. After eating githeri, glucose rises, insulin is released and levels fall again. Pedagogical note: Misconception to correct is that insulin and glucagon are the same thing or that feedback means 'more of the same'. Students add feedback arrows to their models and place any unresolved questions on the DQB, such as how body temperature is controlled.
Lesson 6 Keeping Cool and Staying Warm: Thermoregulation in the Dog, the Lizard and Us	Students looked at temperature data and graphs for a dog, a lizard and a human during a day, and watched or read about behaviours such as panting, basking, shivering, sweating and hiding in burrows. They may have measured their own skin temperature before and after exercise using a thermometer. Teacher note: Ask 'Why does the lizard's temperature change with the rock but the dog's does not?' and give 5 seconds of WAIT TIME. Cold-call 4 students and compare patterns on the board.	Body temperature is kept within a narrow range by negative feedback: thermoreceptors in the skin and hypothalamus detect a change, the hypothalamus acts as the control centre, and effectors respond. When hot, sweat glands release sweat, skin arterioles dilate (vasodilation) so more heat is lost, and hair lies flat; when cold, shivering produces heat, arterioles constrict (vasoconstriction), and hairs stand up. Endotherms (dogs, humans, goats) use internal heat and need much food energy, while ectotherms (lizards) rely mainly on behaviour such as basking and seeking shade. Dogs pant because they have few sweat glands. Teacher note: Emphasise that arterioles do not move closer to the skin; blood flow changes.	Model answer: Different animals use different strategies to keep a steady temperature. Endotherms regulate internally and ectotherms use behaviour, and both have trade-offs between energy use and flexibility. On the same hot afternoon, the dog pants, the lizard moves into the shade, and a person sweats, and each response lowers or prevents a rise in body temperature. Pedagogical note: Students often say that animals 'feel' cold and so shiver, so link to the hypothalamus. Update the class model with the temperature loop and compare the three animals in a table.
Lesson 7 Putting It All Together: Final Explanation of How Animals Keep a Balanced Internal Environment and Remove Toxic Wastes	Students returned to the opening phenomenon of the dog, lizard, tilapia, goat and person on a hot afternoon, and reviewed their DQB, their data from the earlier lessons and the early predictions. They worked in groups to revise the class model and then wrote or presented a final explanation. Teacher note: Give 5 seconds	Homeostasis keeps temperature, water, salts and blood sugar steady using negative feedback, and excretion removes toxic metabolic wastes (ammonia, urea, uric acid) through organs such as the kidneys, liver, skin and lungs. Different animals use different strategies depending on their environment: ammonia in freshwater fish,	Model answer: Animals maintain a balanced internal environment through sensors, control centres and effectors that detect and reverse changes, and they remove toxic wastes in forms suited to their water supply. The dog pants, the lizard basks or hides, the tilapia loses ammonia into water, and the goat and human filter

	of WAIT TIME after asking 'What has changed in your thinking since Lesson 1?' and cold-call 4 students from different groups. Mark the DQB questions that have been answered and note those still open.	urea in mammals, uric acid in water-conserving animals; the kidney filters blood and reabsorbs useful substances; ADH, insulin and glucagon control water and glucose; and the hypothalamus and effectors control temperature, with endotherms and ectotherms using different methods. Teacher note: Check that students can use correct vocabulary and link structure to function.	blood in the kidney, so each keeps conditions inside the body steady despite the hot afternoon outside. Pedagogical note: A strong answer uses evidence from at least three lessons, includes a feedback loop, and compares at least two animals. Use the model and DQB to assess understanding, and address any lingering misconceptions, for example confusing excretion with egestion or thinking that homeostasis means that nothing in the body ever changes.
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